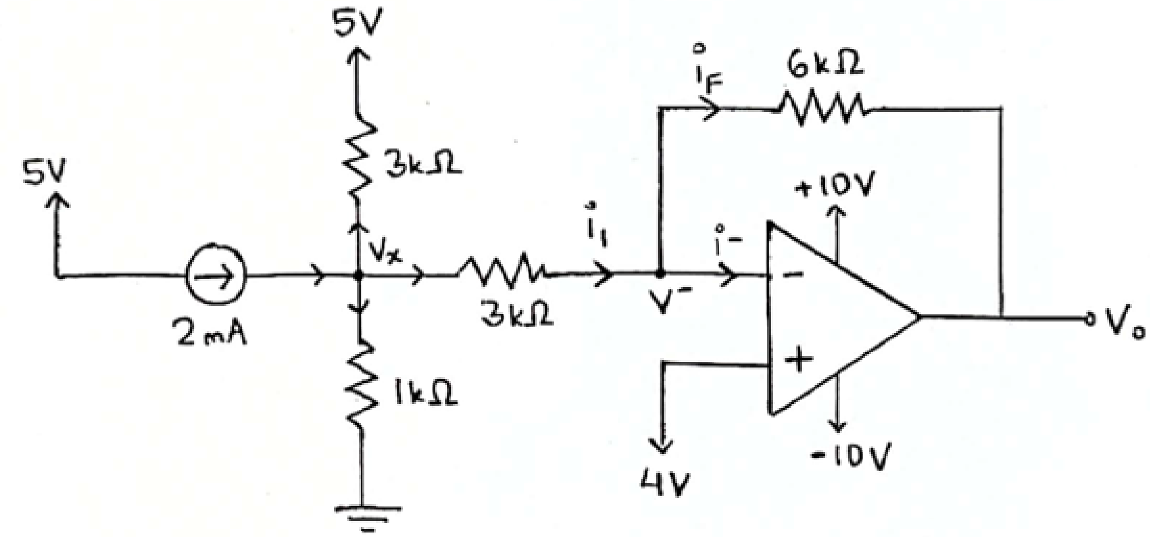
CSE251 Mid Fall 25 (set A)

1) a) $V^- = V^+ = 4V$ (virtual short circuit)

b)



<u>KCL at V_x:</u> $\frac{V_x - 5}{3} + \frac{V_x - V^-}{3} + \frac{V_x}{1} = 2 \quad [V^- = 4V]$

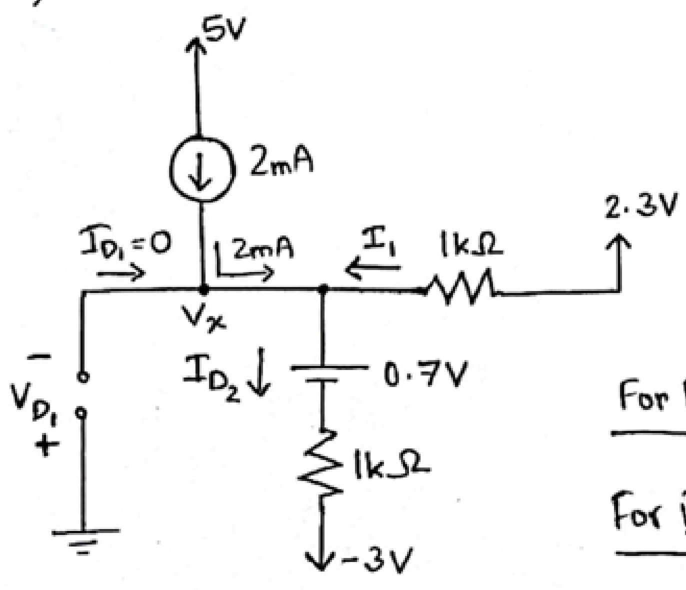
$\Rightarrow V_x = 3V$

c) <u>KCL at V^-:</u> $i_1 = i_F + i^- \quad [i^- = 0]$

$\Rightarrow \frac{V_x - V^-}{3} = \frac{V^- - V_o}{6}$

$\Rightarrow V_o = 6V$

2) Reasonable to assume D_1 is OFF and D_2 is ON.



<u>KCL at V_x:</u>

$2 + I_1 = I_{D_2}$

$\Rightarrow 2 + \frac{2.3 - V_x}{1} = \frac{V_x - 0.7 + 3}{1}$

$\Rightarrow V_x = 1\ V$

Power dissipated in D_2:

$P_{D_2} = I_{D_2} \times V_{D_2}$

$= 3.3mA \times 0.7V$

$= 2.31\ mW$

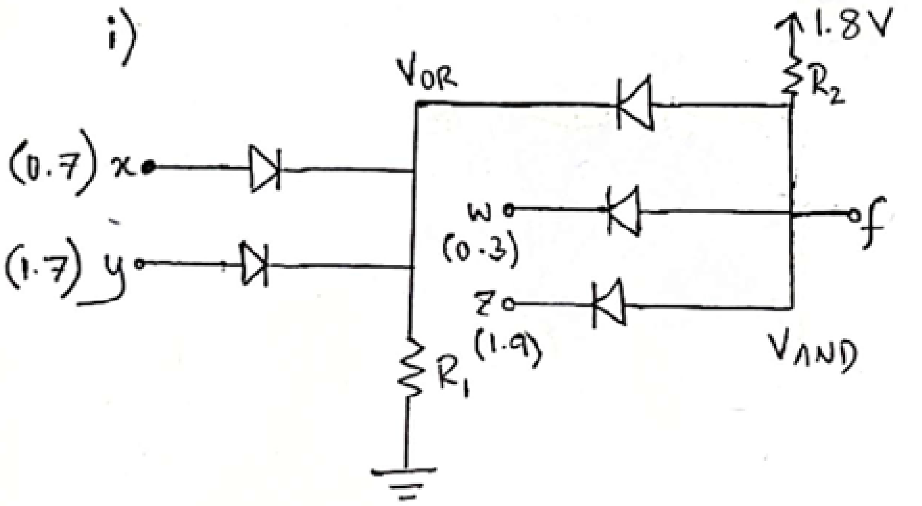
<u>For D_1</u> - $V_{D_1} = 0 - 1 = -1 < 0.7V$ (verified)

<u>For D_2</u> - $I_{D_2} = \frac{1 - 0.7 + 3}{1} = 3.3\ mA > 0$ (verified)

3) a) $f = w(x+y)z \longrightarrow x+y \longrightarrow OR$

$w \cdot (x+y) \cdot z \longrightarrow AND$

i)



ii) $V_{OR} = MAX(0.7-0.7, 1.7-0.7)$
$= 1V$

$V_{AND} = MIN(1+0.7, 0.3+0.7, 1.9+0.7)$
$= 1V$

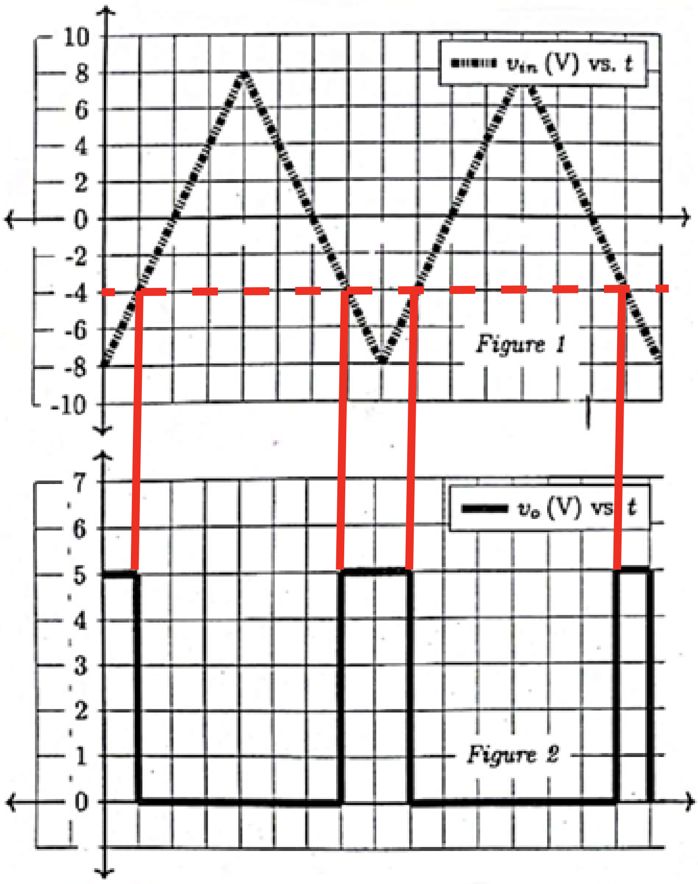
$\therefore f = 1V$

b) Using graph to find the points where switching occurs will give:

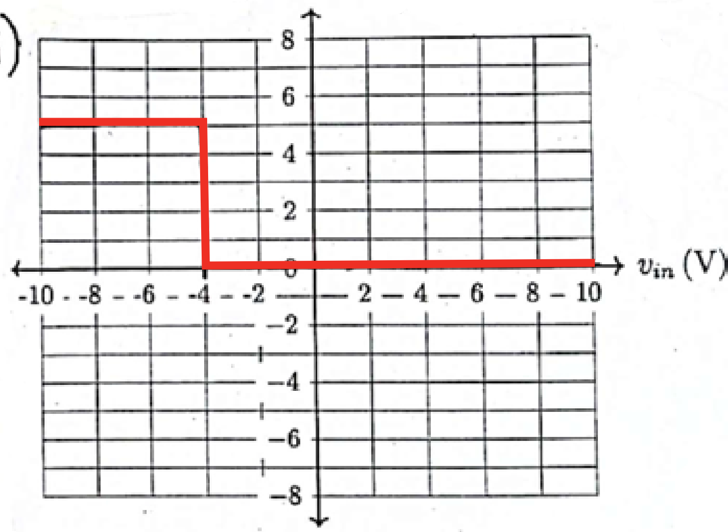
i) $V_{ref} = -4V$

$+V_{sat} = 5V$

$-V_{sat} = 0V$



ii)



4) a) z_1 => output of a differentiator

$$z_1 = -RC\frac{dx}{dt}$$
$$= -100\,\mu F \times 20\,k\Omega\,\frac{dx}{dt}$$
$$= -2\frac{dx}{dt}$$

z_2 => output of a non-inverting amplifier with voltage dividing network at the '+' input ($v+$).

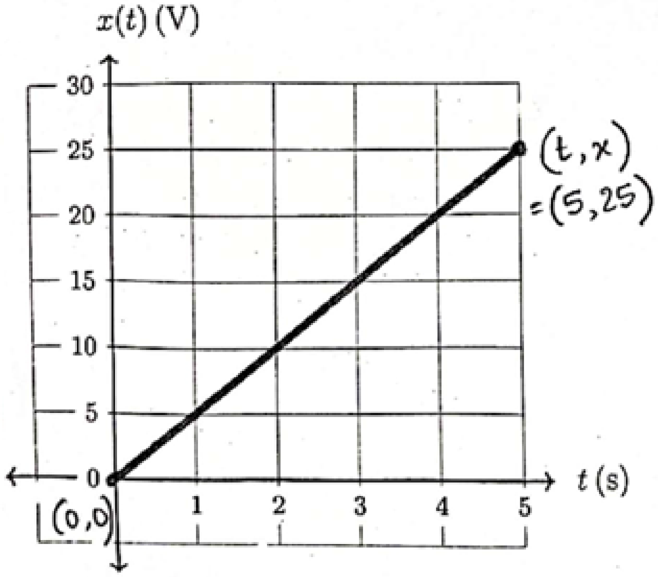
First, $v^+ = \frac{4}{4+8}\times y = \frac{1}{3}y$

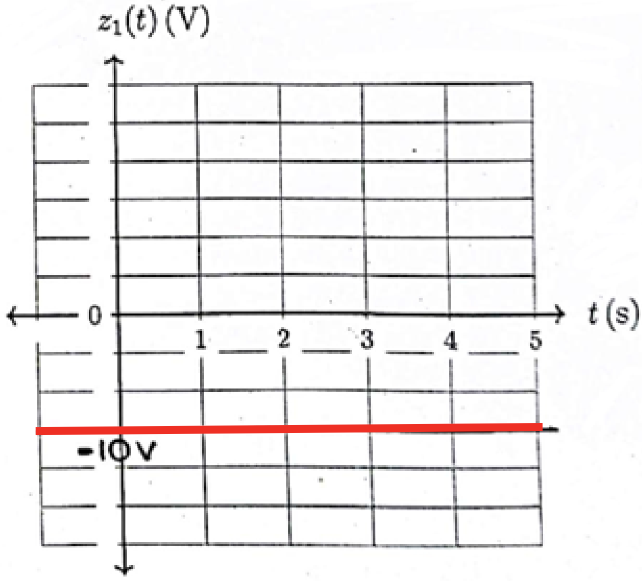
$\therefore\ z_2 = \left(1+\frac{R_F}{R_1}\right)v^+ = \left(1+\frac{1}{2}\right)\frac{1}{3}y = \frac{1}{2}y$

f => output of an inverting summing amplifier with z_1, z_2 as inputs.

$$f = -\frac{1}{1}z_1 - \frac{1}{3}z_2 = +2\frac{dx}{dt} - \frac{1}{6}y$$

b)





slope $= \frac{\Delta x}{\Delta t} = \frac{25-0}{5-0} = 5$

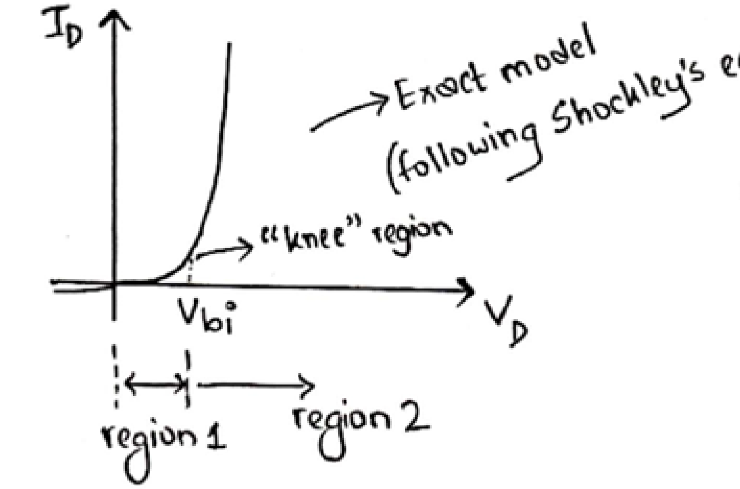
$\downarrow$

$\frac{\Delta y}{\Delta x}$

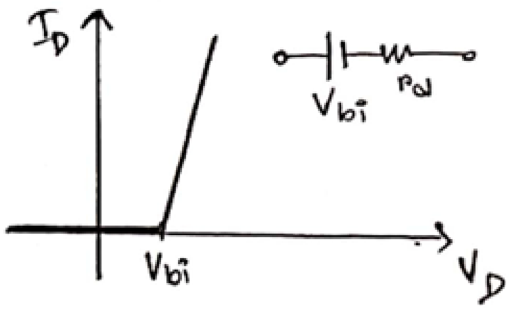
$\therefore$ equation of straight line => $x = 5t$

$\Rightarrow z_1 = -2\frac{dx}{dt} = -2\frac{d}{dt}(5t) = -2\times 5 = -10$

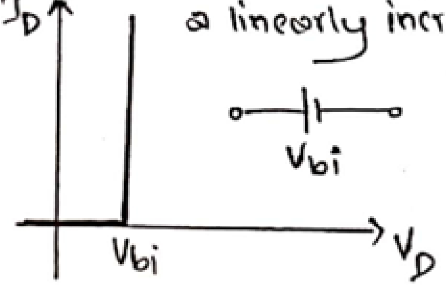
5) a) CVD+R model is more accurate.



* CVD+R model represents both regions 1 and 2 of the exact/real model, with the inaccuracy being the elimination of the "knee" in the curve, and that the exponential increase in region 2 is replaced with a linearly increasing current.



CVD+R



CVD

* CVD model further decreases accuracy, as the increase in voltage in region 2 is neglected (as diode resistance is removed).

b) i) correct

ii) False. It is zero only when negative feedback is provided.

iii) correct

c) From left to right : B, D, D, B